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Multimedia technologies to teach Sign Language in a written form

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Abstract *– In this paper we investigate the features of signs in the Sign Languages and the related problems of how to describe the features linguistically. We present a brief overview of the existing Sign notations; the notations allow us to record all Sign elements in the form of a sequence of characters using alphabetic, digital, and various graphic elements. Also, we substantiate our choice of SignWriting system to record Russian Sign Language, where the signs and rules allow us to set the spatio-temporal form of Signs compactly and precisely. Next, we give examples of a Sign Writing using these characters. Finally, we demonstrate the multimedia program «SWiSL», an on-line platform to help people improve their recognition skills in Sign Language writing.*

Keywords *– Signs; Sign Language; Sign Notation SignWriting; corpus of Signs of Russian Sign Language; distance learning; multimedia tutorial.*

I. Introduction

The multimedia technology has developed and spread enormously in the past decade and drastically changed the ways we conduct various studies and communicate with each other all over the world.

More and more often multimedia technologies are used in education to allow a large number of people to get the training they need via the Internet.

When we develop teaching technologies and tools, it is necessary to take into account the special needs of people. For example, people with auditory pathology have a lot of such needs. Such people are unable to fully perceive speech sounds in a natural way and have to use signs to compensate. In this regard, it seems necessary and important to investigate how to perceive Sign Language and to create means to help such people to communicate.

The aim of this work is to integrate multimedia technologies into a language teaching system. As an example, we present a system to teach native speakers of Sign Language (SL) to use a SL's written form, that is based on SignWriting (SW) Sign notation.

This Sign writing system allows you to accurately capture Signs of any national Sign Language in a static form, to demonstrate the structure of Signs, to identify the features of sign formation, to highlight the signs' general and distinctive features.

The multimedia program «SWiSL» has been developed to remotely train users on-line to perceive written Sign Language. The knowledge gained may be useful for specialists in linguistics and to develop machine Sign recognition systems.

II. Signs and their written form

Speech and Gesture are the most common ways to communicate [1].

Sign language can be considered a variant of a non-verbal communication system, i.e. ordinary gesticulation. In contrast to gesticulation, the components of a Sign in Sign Language are functionally equivalent to phonemes in verbal languages [1, 2].

When a person uses a Sign Language and articulates Signs with spatio-temporal form, he simultaneously express their grammatical features with his hands, the upper part of his body, his head and his face.

Almost every word in Sign Language corresponds to a Sign. Similarly to words, Signs of Sign Language can have several different meanings and be unambiguous, ambiguous, or equivalent. Often, one can fully understand the correct meaning of a Sign only from the context.

When people communicate verbally or via a Sign language, information transmitted from person to person often changes its meaning. The reason for this may be poor-quality speech or fuzzy Signs. At the same time, in speech it is possible to replace a word or Sign with other equivalent words or Signs that sound similarly or have similar components, respectively.

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a word or Sign with other equivalent words or Signs that sound similarly or have similar components, respectively [3].

A more reliable way to preserve the meaning of the transmitted information is to write it down, for example, as Signs.

A. Sign Notations

Sign communication until recently did not have its own written form. This is due to kinetic nature of this kind of communication; such problem, in turn, leads to a set of problems to develop Sign writing systems.

In recent years, several Sign writing systems have appeared. These systems are based on the works of the American linguist W. Stokoe [4].

A review of Sign notations made it possible to distinguish among them the most famous systems in Russia: HamNoSys [5]; V. Sutton - SignWriting (SW) [6]; G. Zaitseva [2] and L. Dimskis [7].

Each of these Sign writing systems using alphabetic, digital and graphic elements and has a number of advantages and disadvantages. Which system to choose depends on the task [8–10].

A comparative analysis of these Sign notations in accordance with their characteristics is based on the selected quality indicators: the detail, the compactness of writing, the visibility, the number of characters used, the presence of facial characters, the versatility [11]. To calculate the indicators, the method of the weighted sum of unequal comparison indicators was used, where all the compared indicators have different importance according to their quantitative estimates obtained via the nominated translation scale [12].

According to the data obtained, the best Sign notations to use is SW system. This SW system allows you to get the record most closely corresponding to the Sign (Fig. 1) [11].

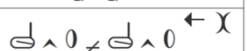
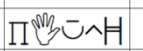
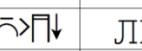
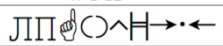
System	The terms and sign the form gestures of sign language		
	Five	*Person*	*Meeting*
Stokoe	5	$\dot{X}_{\perp}^{v}$	$G^{>}G^{>x}$
HamNoSys			
Zayceva	*n* 5 H N	*n* ∋ H ↓	*n* 1 H ← *n* 1 H →
Dimskis			
SW			

Fig. 1. Examples of Signs descriptions of Russian SL with the notation system

It should be noted that it is important to have detailed Sign records, since completely different Signs can include components with minor differences. Insufficiently precise form of Signs can negatively impact how unambiguously we understand the signs.

B. SignWriting notation

One of the features of the SW system is that each component of a Sign in the sign form corresponds to a certain identifier in the form of a code chain [13]. According to the encoding developed by S. Slevinski, complete information about each character can be obtained (Fig. 2).

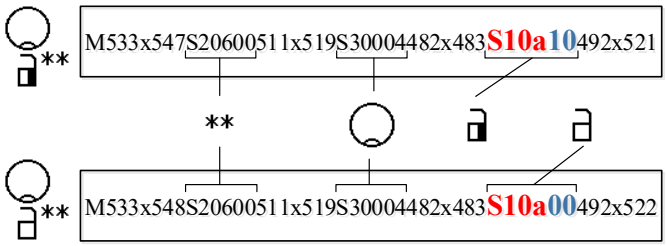


Fig 2. Encoding of Sign Language with similar execution

If we study the code records of these Signs, we will easily identify the sign's distinctive features.

We research Sign Language with the established corpus of the most common gestures of Russian Sign Language (RSL) of various subjects (family, people, information technology, etc.). All elements of the corpus (dictionary) are presented in a written form using the signs of the SW system [14].

The dictionary is the first project in Russia to document Signs of RSL based on the SW system. The dictionary is a practical tool for “read” Signs in a written form.

C. Summary

The Sign recording system SW gives a chance to cover a large number of Signs and to accumulate them in a database. The database can be constantly updated with new elements, various options to apply to certain concepts. We can save this data for as long as we like.

As a student practices a Sign Language with the SW system, the student becomes more proficient with the Sign Language.

A written form of the language would provide the means to improve Sign Language literacy of the speaker.

As you understand individual Signs, you can recognize more complex elements of the Sign Language, like phrases, sentences, complete phrases, and distinguish between similar Signs.

If you know the specifics of how Signs are articulated, you will better understand the signs in general. This helps to eliminate ambiguity and to overcome difficulties in translating Sign Language.

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III. MULTIMEDIA TEACHING PROGRAM

Distance learning to read Sign in sign form in the on-line.

The modern information technology opens new opportunities to educate people. As these technologies are introduced into the traditional education, new teaching methodologies emerge [15].

Multimedia technologies are widely used to intensify education process [11, 12].

Computer-based training systems provide a large number of users with a variety of demonstration and practical materials. These systems also proctor students on remote exam.

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Recently, remote learning, despite rather vague requirements for its quality, has become popular in various fields of study. Remote learning provides a wide range of users with fairly easy and natural interaction with educational materials [16].

For those who want to better understand Sign Language in a written form, a training program «SWiSL» is developed.

This program lets a student identify gesture signs as the student views the corresponding hand-articulated signs.

The exercises offered by the program are designed for the student not to just guess the right sign, but require him to focus and think carefully.

The user can choose between two mods of the program: the demonstration and operational ones. In both mods, the program accompanies every sign with the corresponding text, photo, and video commentaries. If a sign is hard to recognize, this information helps the user to focus on the subtle peculiarities of the sign.

The demonstration mode is exploratory in nature and allows the user to view individual Sign elements (a dactyl (Finger Letter)) and their corresponding sign forms (Fig. 3).

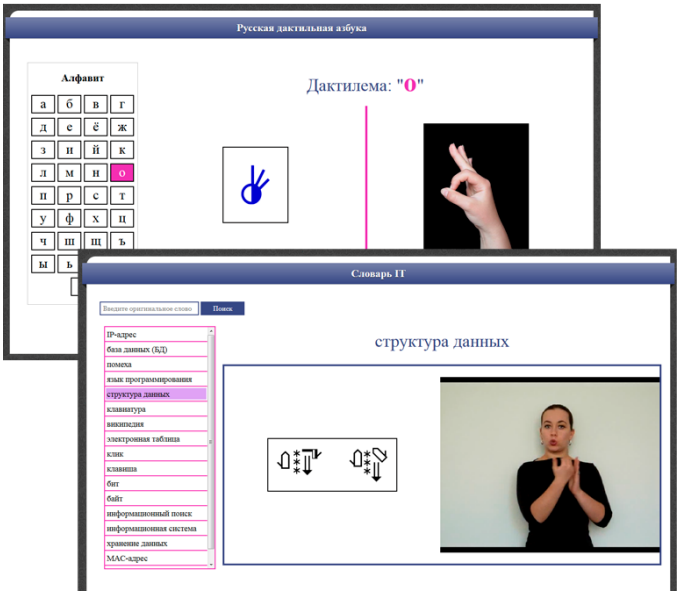


Fig. 3. A fragments of the screen form of the «SWiSL» program in the demo mode

The search system, in response to a user's request, allows the user to view the Sign that corresponds to the sign's iconic analogue and video material.

For the user to search words faster, it's enough to type in the initial letters in the search box. This will reduce the time to select the necessary information from the new list of words displayed on the screen.

In the SW system, words and their components in Sign Language can be recorded by the person performing the gesture, as if he sees himself in a mirror or from the observer side. This explains the mirror image and color filling of the characters displayed on the monitor.

The operating mode of the program, the student solves tests of various difficulty. The tests differ by the time it takes to complete the task. It is possible to use the tooltip and to view the video material.

This program works as follows. The student compares two or more Sign elements in an iconic form and chooses among them the corresponding Sign name. A careful examination of the "signs" presented by the program allows the user to learn to identify nuances. In turn, the user improves his recognition skills.

In testing mode, the user can view iconic Sign forms stored in the database; the forms correspond to individual words, sorted by certain criteria (topics).

The user is challenged to determine the meaning of a gesture given the visual representation of the gesture - Signs in an iconic form (SignWriting).

The program begins a dialogue with the user in a test mode and shows a sign analogue of a gesture on the screen. One of the characters represented corresponds to the video, and the rest are selected using a random number generator (Fig. 4).

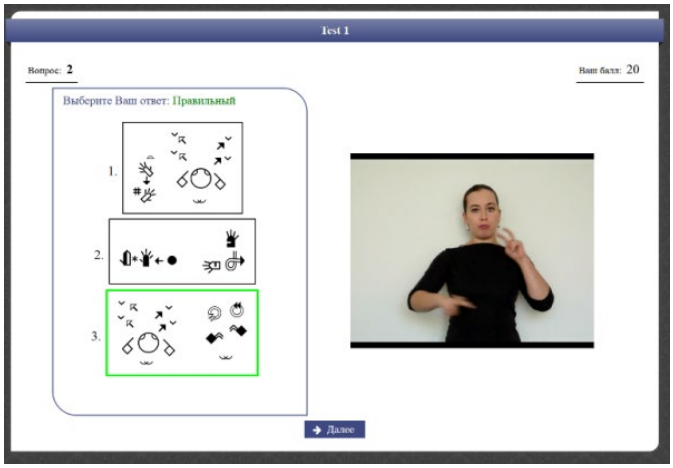


Fig. 4. A screen fragment in the dialogue mode of our program

The program «SWiSL», in addition to its teaching function, can quickly assess the user's results. The program also collects, stores, and processes all user responses in a database. The stored results can be used in further research.

The multimedia program «SWiSL» is self-mastery tool to read Signs in sign form, available anytime and anywhere via the Internet. This program can be a starting point for user to get acquainted with the SW system. Time will tell whether it will be accepted by native RSL users.

The system can be further development with new functionality. For example, a user will be able to view the dictionary in an electronic form with variable Signs.

This will solve certain difficulties in creating systems for automatically recognize Sign speech.

CONCLUSIONS

Recent studies, conducted in the field of Sign Language and how to identify features and specifics of Sign notations, confirm the relevance of our chosen topic.

A qualitatively new direction to introduce multimedia technologies into the educational process is to create multimedia-based training programs.

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The program «SWiSL» teaches how to use Sign notation *SignWriting*. The program allows a user to study signs in detail.

The results can be used by Sign Language specialists and developers of machine translation systems in order to recognize signs more accurately.

The scientific novelty of this work is to integrate multimedia technologies into a single education system; the system teaches Sign Language in a symbolic form.

The practical significance of the work is the multimedia program «SWiSL». The program is a qualitatively new approach to teach Sign Language in a written form.

At the moment, despite all the work done in different countries to study Sign Language, there are still many insufficiently explored areas. Specialists in the field of linguistics, programming, and, most importantly, Sign Language carriers who are well aware of all its rules and subtleties will be highly to teach Sign Language in a written form.

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